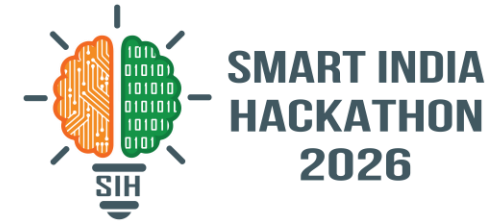


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26038**
- **Problem Statement Title-** Explainable AI for Diabetic Retinopathy Screening in Rural India
- **Theme-** Clean & Green Technology
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)**



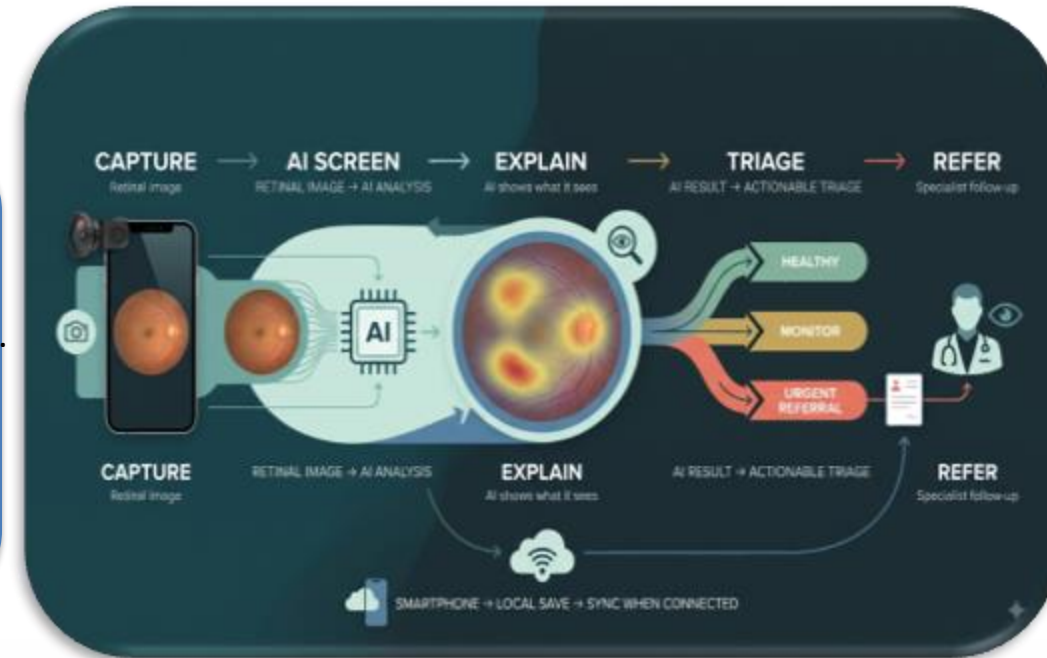
"The eyes are the window to your health, not just your soul."

❖ Problem Statement:

- **Silent Onset:** Diabetic retinopathy **shows zero symptoms** until vision damage has already begun.
- **Limited Access:** Rural India has almost **no ophthalmologists** within reach for early screening.
- **Disconnected Care:** No link between screening, diagnosis, and specialist follow-up at the village level.
- For **India's 77M+ diabetics**, diabetic retinopathy remains a leading *preventable* cause of blindness — yet screening requires a retinal specialist and fundus camera, both scarce outside tier-1 cities, so patients go undiagnosed until vision loss is already severe and irreversible.

❖ Solution:

- **Smartphone-First Capture:** Just a phone and a low-cost clip-on retinal lens, no expensive fundus camera or specialist hardware needed.
- **On-Device AI Detection:** A fine-tuned **EfficientNet-B0 model** grades DR severity instantly, right on the device.
- **Explain, Don't Just Predict:** A **Grad-CAM heatmap** overlays the exact region driving the AI's verdict, so the health worker sees and trusts the reasoning.
- **Triage, Not Diagnose:** Results sort into **Healthy, Monitor, or Urgent Referral**: safe, actionable, never a false diagnosis.
- **Built for the Field:** Runs **fully offline** on-device, syncing to the cloud only when signal returns.



INNOVATION & UNIQUENESS



SELF-EXPLAINING AI

Most tools give a verdict and expect trust. Ours **shows its reasoning** directly **on the image**, so the health worker can actually see what the AI saw.



TRIAGE, NOT DIAGNOSIS

We never claim to diagnose. Sorting patients into **Healthy**, **Monitor**, or **Urgent Referral** keeps the tool safe, honest, and genuinely useful in the field.



OFFLINE-FIRST BY DESIGN

Built for the reality of rural connectivity, not against it. A full day's screenings **run and save locally**, **syncing only when signal returns**.
SCREEN → SAVE → SYNC
LOCALLY → LATER



ZERO EXTRA HARDWARE

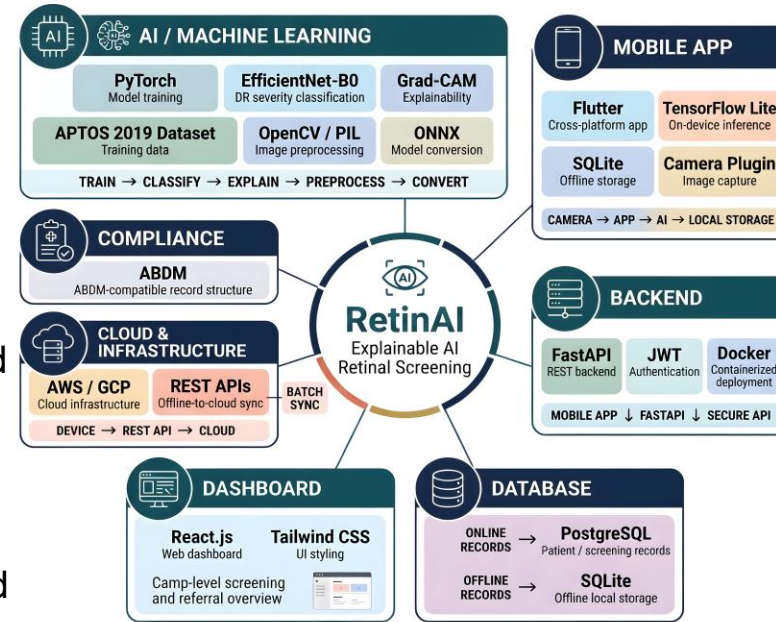
No custom scanner, no proprietary device. It works on any smartphone camera a health worker already carries.

"Somewhere in rural India, someone is losing their sight to a disease that takes five minutes to catch."



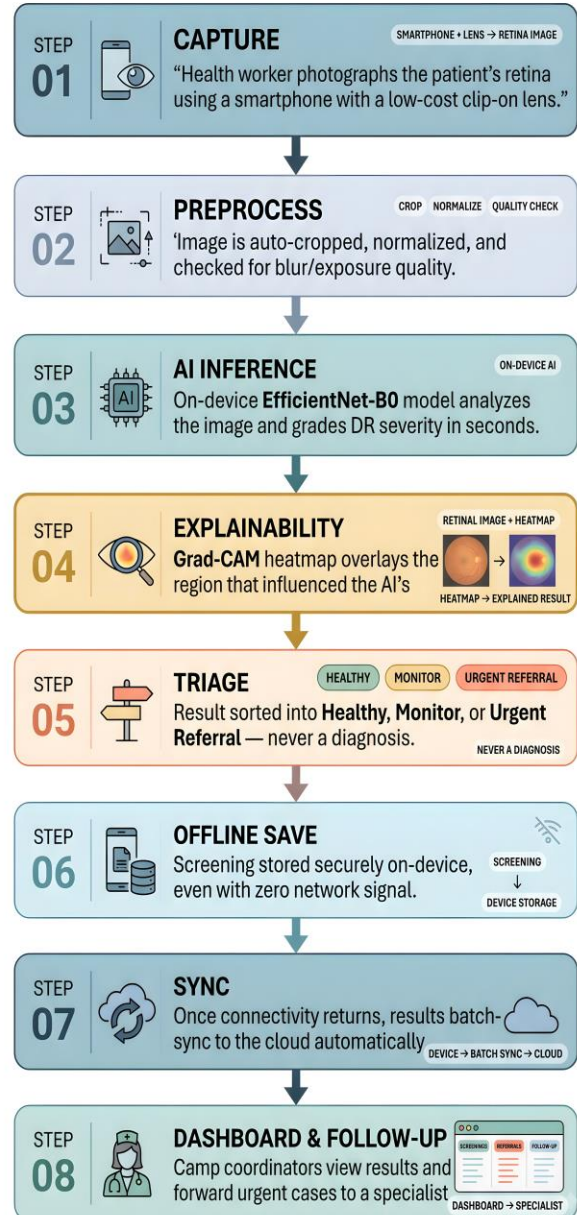
TECHNOLOGIES USED

Technology stack powering RetinAI



AI → MOBILE → BACKEND → CLOUD → DASHBOARD
Complete technology stack for explainable, offline-first retinal screening.

WORKFLOW

❖ **TECHNOLOGIES USED:**

- **AI / Machine Learning:** PyTorch, EfficientNet-B0 (DR severity classification), Grad-CAM (explainability), APTOS 2019 dataset (training data), OpenCV/PIL (preprocessing), ONNX (model conversion)
- **Mobile App:** Flutter, TensorFlow Lite (on-device inference), SQLite (offline storage), Camera plugin (image capture)
- **Backend:** FastAPI, JWT (authentication), Docker (containerized deployment)
- **Database:** PostgreSQL (patient/screening records), SQLite (offline local storage)
- **Dashboard:** React.js, Tailwind CSS (UI styling)
- **Cloud & Infrastructure:** AWS/GCP, REST APIs (offline-to-cloud sync)
- **Compliance:** ABDM-compatible record structure

❖ **Methodology (Step-by-Step):**

- **Capture** → Smartphone + clip-on lens photographs the patient's retina.
- **AI Analysis** → On-device model grades DR severity and generates a Grad-CAM heatmap for explainability.
- **Triage** → Result sorted into Healthy, Monitor, or Urgent Referral.
- **Sync & Dashboard** → Screening saved offline, synced to cloud when signal returns, visible on coordinator dashboard.
- **Follow-up** → Urgent cases forwarded to a specialist for confirmation.

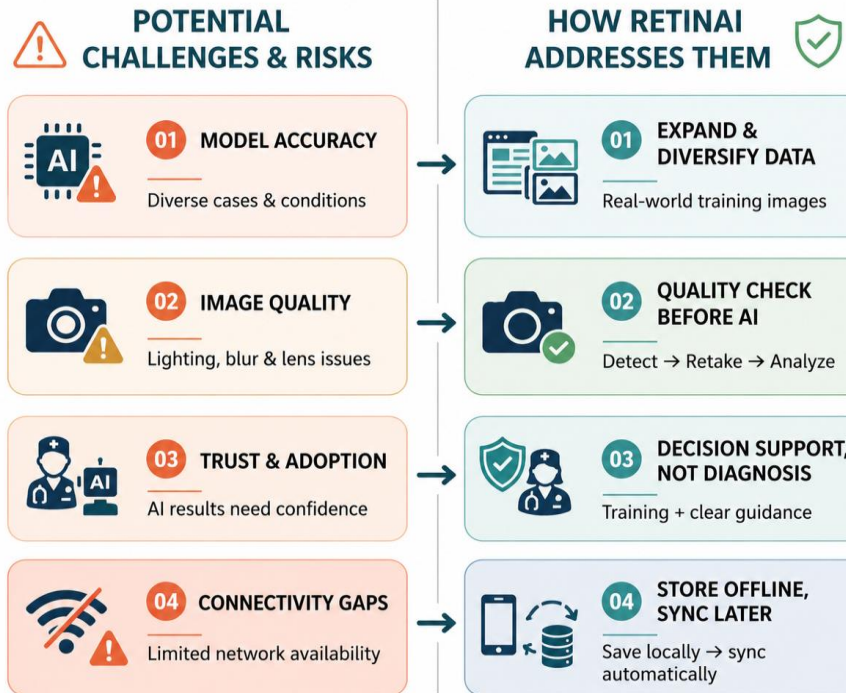
"The best technology doesn't replace doctors — it reaches the people doctors can't."

❖ Feasibility:

- Retinal capture components (clip-on lens, smartphone camera) are commercially available in India.
- EfficientNet-B0 + public APTOS dataset → retrainable and fine-tunable for our specific triage use case.
- Grad-CAM explainability already proven in medical imaging research.
- MVP in 36 hours: capture module, on-device AI inference, Grad-CAM overlay, basic triage dashboard.

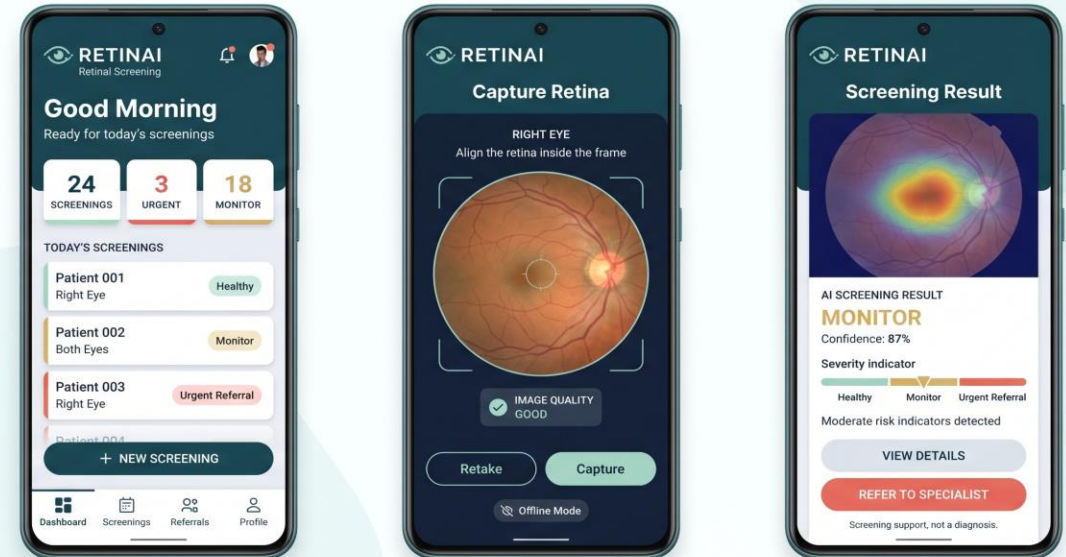
❖ Viability:

- Deployment: Pilot in one health camp, scalable to hundreds of patients across districts.
- Cost: Single smartphone + lens setup replaces the need for a dedicated fundus camera per site.
- Adoption: Minimal training needed for health workers, automatic triage logs → high acceptance by camp coordinators.



ANDROID APP — DASHBOARD

RETINAI mobile screening interface



❖ **Who Benefits:**

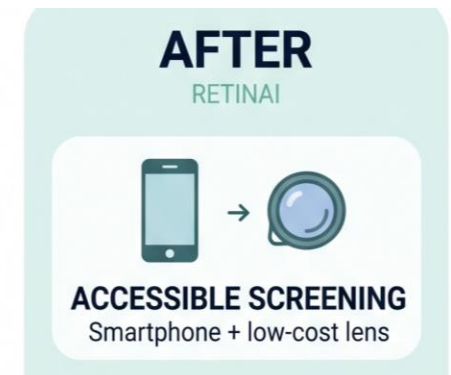
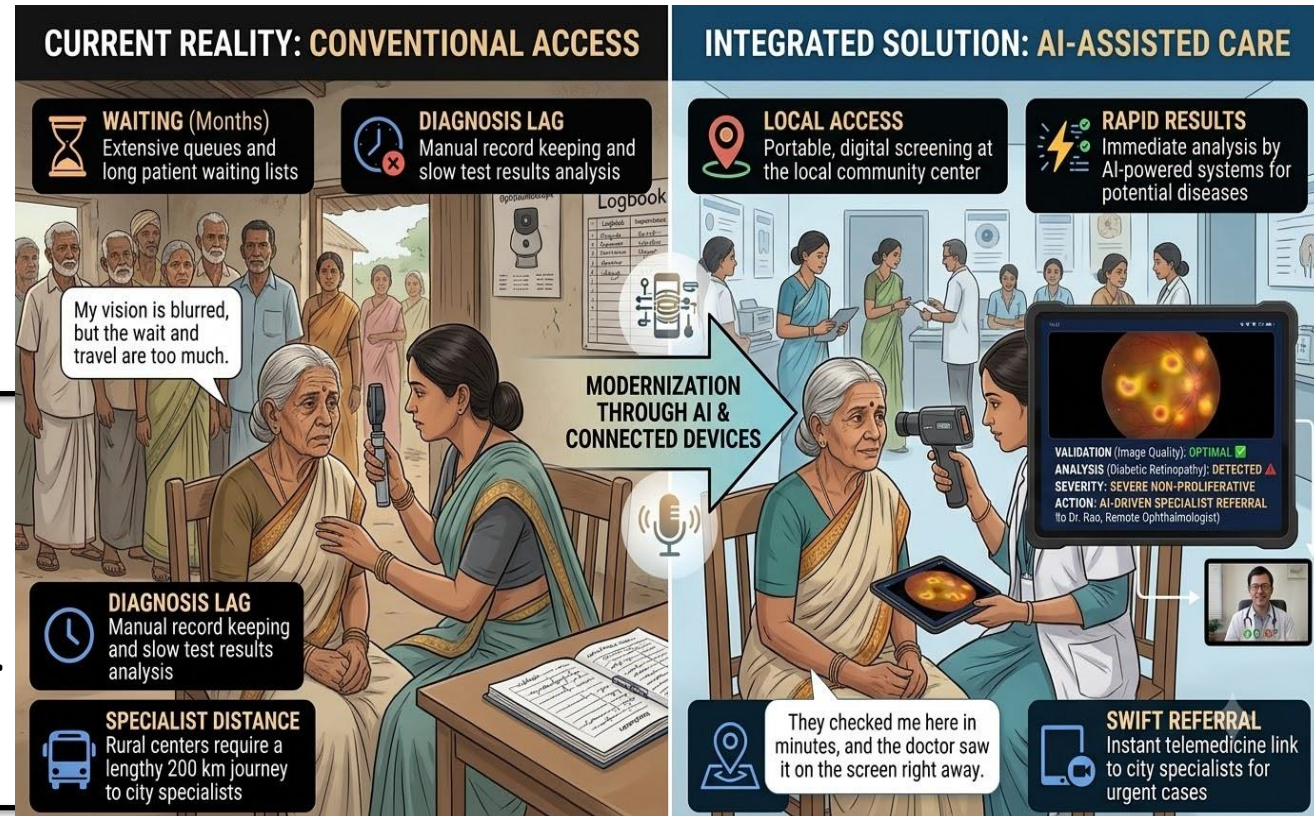
- Rural diabetic patients with no nearby specialist access.
- ASHA/health workers running diabetes screening camps.
- NGOs and state health departments running rural outreach.
- Ophthalmologists, via smarter, pre-filtered referrals.

❖ **Direct Benefits:**

- Catches DR before symptoms appear → prevents avoidable blindness.
- Screening in minutes, no travel to a specialist required.
- Low-cost, smartphone-based → no expensive equipment needed.
- Works fully offline → built for real rural connectivity.
- Explainable results → builds health-worker trust in the AI.

❖ **Alignment:**

- Ayushman Bharat Digital Mission (ABDM) → digital health record push.
- Digital India → AI-driven healthcare access at the last mile.
- National Programme for Control of Blindness → preventable-blindness reduction goals.
- Sustainable Development Goals (SDG 3) → good health & well-being for all.



"The only thing worse than being blind is having sight but no vision."

RESEARCH AND REFERENCES:

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2. **APTOS 2019** – Blindness Detection Dataset
<https://www.kaggle.com/competitions/aptos2019-blindness-detection/data>
3. **EfficientNet** – CNN Architecture
<https://arxiv.org/abs/1905.11946>
4. **Grad-CAM** – Explainable AI
<https://arxiv.org/abs/1610.02391>
5. **AI-Based Diabetic Retinopathy Screening**
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7307769/>
6. **Indian DR Screening Guidelines** – RSSDI & VRSI
<https://link.springer.com/article/10.1007/s13410-023-01296-z>

OUR PROJECT REPOSITORY:

- ✓ Research Papers & References
- ✓ High-quality Model Images & Flowcharts
- ✓ Future Scope Concept



Scan me!